INCH POUND

A-A-52023
February 19, 1991

COMMERCIAL ITEM DESCRIPTION

TARGET SET, SURVEYING

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-T-52823.

This commercial item description contains the functional, performance, and design characteristics required of a target set used as a sighting object for theodolites in the conduct of field surveys (i.e. traverse, triangulation, azimuth observations) at accuracies of Second Order or lower. It is essential that the target set has tribrach mounts compatible with the theodolites used to accommodate standard surveying field practices (i.e. "leap-frogging"). The target set shall consist of the following major components:

Item	Quantity
Target assembly	1
Tribrach assembly	1
Battery Box	1
Transport case	1

Salient characteristics:

Target assembly: The target assembly shall consist of a target plate of frosted translucent glass/plexiglass mounted inside a corrosion-resistant metal (CRM) frame approximately 4.5 inches (115 mm) square and 0.35 inches (9 mm) deep, flat black finish. A centering mark shall be located at the exact center of the top rim of the frame and shall be finished in gloss white or other highly reflective material. The target plate shall be masked with triangular cut-outs arranged to provide a clearly visible sighting pattern for both horizontal and vertical angular measurements at a range of 0.6 mile (1 kilometer) when viewed through a theodolite during

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AMSC N/A FSC 6675

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daylight hours under nominal atmospheric conditions. The target frame/plate component of CRM shall be mounted on a vertical standard or carrier which will have a base containing clutch bolts designed to mate with the universal tribrach lever or cam-action locking device. The carrier shall contain a level housing and be capable of independent rotation about the vertical axis 360 degrees in both clockwise and counter-clockwise directions. The level housing portion of the carrier shall contain a liquid-filled tubular level vial, marked with permanent graduations at 2mm intervals with a sensitivity of approximately 45 to 75 arc-seconds per graduation. The level vial housing shall be fitted with a mechanism for independent adjustment of the enclosed vial. A means of background illumination of the translucent target plate shall be provided using incandescent lighting powered by either an integrated small battery source or separate tripod mounted battery pack.

Tribrach assembly: The tribrach is a mounting/leveling/centering mechanism which accommodates interchangeably the base assemblies of the target and the theodolite used in the conduct of field operations. The tribrach shall be diecast of corrosion-resistant alloy and be configured with three integral flanges spaced 120 degrees apart. The finish shall visually approximate green 383. The visual matching can be achieved if item is painted in accordance with TT-E-529, matching color chip 24102. Each flange shall accommodate adjustable leveling screws from material conforming to ASTM B 151, UNS 79200, H01 temper. The adjusting nuts and milled leveling knobs shall conform to ASTM B 16, H02 temper. The leveling screws shall permit leveling of the target when the base plate is not more than 6 degrees out of level. The base of the tribrach shall contain three clutch bolt holes to allow mating with target base or theodolite clutch bolts. The holes shall be centered on the same 120 degree tribrach radials and 1.25 inches (31.75 mm) from the tribrach center point. The tribrach will be connected to a base plate which provides a bearing surface of corrosion-resistant case hardened steel for mounting the tribrach on the tripod head. The center of the base plate shall have a 5/8-11 UNC-2B nut for mating to the tripod head. A means of providing equal downward tension perpendicular to each radial arm of the base plate shall be accomplished using a tempered plate spring, either plated or treated to resist corrosion and attached to the base plate with a nut and lock screw. A circular level shall be in a metal housing as part of one of the tribrach flange castings. The level contained therein shall be adjustable and have a sensitivity of between 7-13 arc-minutes per 2mm graduation. The center of the vial shall be etched with a circle 2mm in diameter. A means of rapidly disengaging the target/theodolite base from the tribrach assembly shall be provided by a locking device controlled by a lever or cam-action knob composed of material conforming to ASTM B 584, UNS C86400 or C86700. The locking plate motion shall secure the clutch bolts of the target base with an error of centering not to exceed 0.002 inch (0.5 mm). An optical plumbing device shall be affixed to the tribrach. The optics shall provide a clear image of the reticle and objective target throughout an operating range of 3 feet (0.914m) to not less than 110 feet (33.5m). The reticle shall be

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adjustable to true plumb and provide a positive centering pattern such as two mutually perpendicular etched lines intersecting at the center of the field of view.

Battery box: A battery box/case shall be provided to separately house commercially available dry cell batteries required as a power source for the illumination component of the target. Materials for the case may be of CRM, fiberglass-reinforced plastic laminate or high impact plastic capable of withstanding temperature ranges, shock/vibration conditions and moisture encountered in a field surveying environment annually worldwide. The finish shall visually approximate green 383. The visual matching can be achieved if item is painted in accordance with TT-E-529 matching color chip 24102. Sufficient power shall be provided to support illumination of the target for not less than 6 hours of continuous operation by sustaining a visible target pattern when viewed by a theodolite at a range of 0.6 mile (1 kilometer) in nominal atmospheric conditions. A rheostat shall be provided on the case to allow for the selection of light intensity of target illumination. The battery box shall be fitted with a means of attaching the case to a tripod during operation and a power cable of not less than 36 inches (914 mm) in length for connection to the target lamp.

Transport case: A transport/carrying case of plywood, molded thermoplastic resin, fiberglass-reinforced plastic laminate or other materials capable of withstanding the operational environment without damaging the target set contained therein shall be provided. The finish shall visually approximate green 383. The case shall have a carrying strap or handle. All metal parts shall be corrosion resistant. The interior of the case shall be partitioned and padded to allow separation of set components for maximum support and protection against vibration, shock, leakage of water, or exposure to dust encountered in the operational environment.

Workmanship: All parts shall be of first class workmanship and shall be free from defects of manufacture which detract from their appearance or may impair their serviceability.

Regulatory requirements. In accordance with Section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

Certification of commercial item. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices and is the same product as the manufacturer's current product offered by the contractor in the commercial marketplace for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to

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first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric Products. This CID specifies product requirements in inch-pound units. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Preservation, packaging, packing, labeling and marking. Preservation, packaging, packing, labeling and marking shall be as specified in the contract or order.

Notes:

1. Ordering data. Procurement documents should specify the following:

a. Make and model of theodolite to be used with target set.
b. Level of packaging required.
c. When additional marking is required.

2. When this commercial item description is used for procurement, the commercial item certification clause must appear in the solicitation.

3. FED-STD-376 and TT-E-529 are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

4. ASTM B 151, ASTM B 16, and ASTM B 584 are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

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MILITARY INTERESTS:

Custodians:

Army - ME
Air Force - 99

Review activities:

Air Force - 82
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

Preparing activity:
Army - ME

Project 6675-0941

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